Test Cases - Restaurant Billing System (PS 43)

File: restaurant_billing.py | Menu: Dosa 80, Idli 40, Thali 150, Tea 20 | Discount: 10% if subtotal > 500 | GST: 5% on (subtotal - discount)

ID	Scenario	Inputs (in order)	Expected Result	Status
TC1	Normal order, no discount (subtotal <= 500)	Dosa, 2, Tea, 1, bill	Subtotal 180.00, Discount 0.00, GST 9.00, Grand Total 189.00	Pass
TC2	Discount applies (subtotal > 500)	Thali, 4, bill	Subtotal 600.00, Discount 60.00, GST 27.00, Grand Total 567.00	Pass
TC3	Boundary: subtotal exactly 500 (no discount)	Thali, 2, Dosa, 2, Idli, 1, bill	Subtotal 500.00, Discount 0.00, GST 25.00, Grand Total 525.00	Pass
TC4	Just above boundary: subtotal 530	Thali, 3, Dosa, 1, bill	Subtotal 530.00, Discount 53.00, GST 23.85, Grand Total 500.85	Pass
TC5	Same item ordered twice is merged	Idli, 2, Idli, 3, bill	Idli x5 = 200; Subtotal 200.00, GST 10.00, Grand Total 210.00	Pass
TC6	Case-insensitive item name	dosa, 1, bill	Accepted as Dosa; Grand Total 84.00	Pass
TC7	Item not on menu	Pizza, Tea, 1, bill	'Item not on the menu' message, then order continues; Grand Total 21.00	Pass
TC8	Invalid quantity (text)	Dosa, abc, 1, bill	'Invalid quantity' message; re-prompt for item; Dosa then ordered x1	Pass
TC9	Invalid quantity (negative / decimal)	Dosa, -3, Dosa, 2.5, Tea, 1, bill	'Invalid quantity' shown twice; Grand Total 21.00	Pass
TC10	Quantity zero	Idli, 0, Idli, 1, bill	'Quantity must be at least 1.'; Grand Total 42.00	Pass
TC11	Bill requested with empty order	bill, Tea, 1, bill	'No items ordered yet.' then normal bill; Grand Total 21.00	Pass

Formula: Discount = 10% of Subtotal (only if Subtotal > 500); GST = 5% of (Subtotal - Discount); Grand Total = Subtotal - Discount + GST.